

User Stories and Acceptance Criteria

Project Name

CarbonWise

Version: 1.0

Status: Approved

1. Purpose

This document defines the functional requirements of CarbonWise from the user's perspective.

It ensures every feature directly supports the challenge statement:

Design a solution that helps individuals understand, track, and reduce their carbon footprint through simple actions and personalized insights.

The document is also used by:

- Developers
- QA Engineers
- Judges
- Future Contributors

to verify that the system fulfills its intended purpose.

2. User Personas

Persona 1 — Student

Name: Rahul

Age: 20

Characteristics:

- Budget conscious

- Technology friendly
- Interested in sustainability

Goals:

- Understand environmental impact
 - Learn sustainable habits
-

Persona 2 — Working Professional

Name: Priya

Age: 29

Characteristics:

- Limited free time
- Data-driven decision maker

Goals:

- Reduce emissions efficiently
 - Receive personalized advice
-

Persona 3 — Sustainability Enthusiast

Name: Arjun

Age: 32

Characteristics:

- Highly motivated
- Tracks environmental impact

Goals:

- Optimize lifestyle choices
 - Measure long-term improvements
-

3. Epic 1 — Authentication

Supports:

Secure Access

Code Quality

Security

User Story 1.1

As a new user,

I want to create an account,

So that I can access CarbonWise.

Acceptance Criteria

Given:

User is on registration page

When:

Valid information is entered

Then:

Account is created successfully

And:

User can log in immediately

User Story 1.2

As a registered user,

I want to securely log in,

So that I can access my sustainability dashboard.

Acceptance Criteria

Given:

Valid credentials

When:

Login is attempted

Then:

JWT token is issued

And:

Dashboard loads successfully

User Story 1.3

As a user,

I want invalid login attempts rejected,

So that my account remains secure.

Acceptance Criteria

Given:

Incorrect password

When:

Login is attempted

Then:

Access denied

And:

Error message displayed

4. Epic 2 — Carbon Understanding

Supports:

Understand

from the challenge statement.

User Story 2.1

As a user,

I want to calculate my carbon footprint,

So that I understand my environmental impact.

Acceptance Criteria

Given:

Transportation

Food

Energy

Shopping

data is entered.

When:

Calculation is submitted

Then:

Carbon score generated

And:

Emission breakdown displayed

User Story 2.2

As a user,

I want to identify my largest emission source,

So that I know where improvements matter most.

Acceptance Criteria

Given:

Multiple activity categories

When:

Score calculation completes

Then:

Highest emission category highlighted

User Story 2.3

As a user,

I want a sustainability rating,

So that I can quickly understand my performance.

Acceptance Criteria

Ratings available:

Eco Champion

Sustainable

Improving

High Impact

Critical

5. Epic 3 — Progress Tracking

Supports:

Track

from the challenge statement.

User Story 3.1

As a user,

I want to see my carbon history,

So that I can measure progress.

Acceptance Criteria

Dashboard displays:

- Monthly Trends
 - Historical Emissions
 - Carbon Score Changes
-

User Story 3.2

As a user,

I want activity history,

So that I can review previous entries.

Acceptance Criteria

User can:

- Search Activities
 - Filter Activities
 - View Activity Details
-

User Story 3.3

As a user,

I want weekly and monthly summaries,

So that I can understand trends.

Acceptance Criteria

Reports include:

- Total Emissions
 - Category Breakdown
 - Improvement Metrics
-

6. Epic 4 — Personalized Recommendations

Supports:

Reduce

from the challenge statement.

User Story 4.1

As a user,

I want personalized recommendations,

So that I can reduce emissions effectively.

Acceptance Criteria

Recommendations generated using:

- Emission Hotspots
 - User Preferences
 - Historical Activities
-

Example

Transportation-heavy user receives:

Use Public Transport

Food-heavy user receives:

Reduce Meat Consumption

User Story 4.2

As a user,

I want recommendations ranked by impact,

So that I know which action matters most.

Acceptance Criteria

Each recommendation displays:

- Priority
 - Estimated Savings
 - Difficulty Level
-

User Story 4.3

As a user,

I want recommendations tracked,

So that I can measure behavior change.

Acceptance Criteria

System records:

- Accepted Recommendations
 - Completed Recommendations
 - Adoption Rate
-

7. Epic 5 — Sustainability Goals

Supports:

Track

Reduce

User Story 5.1

As a user,

I want to create sustainability goals,

So that I can reduce emissions over time.

Acceptance Criteria

User can:

- Create Goal
- Edit Goal
- Delete Goal

User Story 5.2

As a user,

I want progress tracking,

So that I know how close I am to my target.

Acceptance Criteria

System displays:

- Completion Percentage
- Milestones
- Remaining Reduction

User Story 5.3

As a user,

I want milestone notifications,

So that I stay motivated.

Acceptance Criteria

Notifications sent at:

25%

50%

75%

100%

completion.

8. Epic 6 — Carbon Twin

Supports:

Personalized Insights

User Story 6.1

As a user,

I want to simulate future scenarios,

So that I understand potential outcomes before making changes.

Acceptance Criteria

User can modify:

- Transportation
- Food
- Energy

variables.

Example

Reduce Car Usage

Output:

Projected Emissions

Potential Savings

User Story 6.2

As a user,

I want future carbon forecasts,

So that I can make informed decisions.

Acceptance Criteria

Forecast includes:

- Current Emissions
 - Predicted Emissions
 - Reduction Percentage
-

9. Epic 7 — Accessibility

Supports:

Accessibility (Low Impact)

User Story 7.1

As a keyboard-only user,

I want complete keyboard navigation,

So that I can use all features.

Acceptance Criteria

All pages support:

- Tab Navigation
 - Enter Key
 - Escape Key
-

User Story 7.2

As a screen reader user,

I want meaningful labels,

So that I can understand content.

Acceptance Criteria

All forms include:

- Labels
 - ARIA Attributes
 - Error Announcements
-

User Story 7.3

As a low-vision user,

I want sufficient color contrast,

So that content remains readable.

Acceptance Criteria

Minimum contrast:

4.5:1

10. Epic 8 — Security

Supports:

Security (Medium Impact)

User Story 8.1

As a user,

I want my account protected,

So that unauthorized users cannot access my data.

Acceptance Criteria

System uses:

- JWT Authentication
 - bcrypt Password Hashing
 - Rate Limiting
-

User Story 8.2

As a user,

I want secure data submission,

So that malicious input is rejected.

Acceptance Criteria

Input validation implemented using:

Zod

Invalid Requests

Must return:

400 Bad Request

User Story 8.3

As an administrator,

I want common attacks prevented,

So that the platform remains secure.

Acceptance Criteria

Security protections include:

- Helmet
 - Input Validation
 - Authorization Checks
 - Secure Headers
-

11. Epic 9 — Performance

Supports:

Efficiency (Medium Impact)

User Story 9.1

As a user,

I want fast dashboard loading,

So that the platform feels responsive.

Acceptance Criteria

Dashboard loads:

< 2 Seconds

User Story 9.2

As a user,

I want quick recommendations,

So that insights feel immediate.

Acceptance Criteria

Recommendation generation:

< 500ms

User Story 9.3

As a user,

I want fast carbon calculations,

So that I receive results instantly.

Acceptance Criteria

Calculation time:

< 100ms

12. Epic 10 — Testing and Reliability

Supports:

Testing (Low Impact)

User Story 10.1

As a developer,

I want unit tests,

So that business logic remains reliable.

Acceptance Criteria

Coverage:

90%+

for core engines.

User Story 10.2

As a developer,

I want integration tests,

So that modules interact correctly.

Acceptance Criteria

Coverage:

80%+

for API flows.

User Story 10.3

As a developer,

I want end-to-end tests,

So that critical user journeys remain functional.

Acceptance Criteria

Playwright tests cover:

- Registration
 - Login
 - Calculator
 - Recommendations
 - Goals
 - Carbon Twin
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13. Final Acceptance Criteria

CarbonWise is accepted as complete when:

- ✓ Users can understand their carbon footprint
 - ✓ Users can track sustainability progress
 - ✓ Users receive personalized recommendations
 - ✓ Carbon Twin simulations function correctly
 - ✓ Security protections are implemented
 - ✓ Performance targets are achieved
 - ✓ Accessibility requirements are satisfied
 - ✓ Testing requirements are satisfied
 - ✓ Documentation is complete
 - ✓ Repository follows challenge guidelines
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14. Business Success Criteria

The project is considered successful when:

- ✓ Users identify their emission hotspots
- ✓ Users adopt recommendations
- ✓ Users reduce emissions over time
- ✓ Sustainability awareness increases
- ✓ Carbon footprint tracking becomes habitual
- ✓ Personalized insights drive behavior change
- ✓ The platform clearly fulfills the challenge statement